Model Setting

$$y_t = f(\lambda_t, k_t, n_t, z_t)$$

$$= \lambda_t \cdot n_t^{\theta} \cdot \left(\left[(1-\sigma) k_t^{-\nu} + \sigma \cdot \underbrace{z_t^{-\nu}}_{\text{存货}} \right]^{\frac{-1}{\nu}} \right)^{1-\theta}$$

- k_t / z_t : elasticity $\frac{1}{\nu+1}$
- labor share: θ

$0 < \theta < 1$

$0 < \sigma < 1$

$\nu > 0$

引入资本建造

- J : 共 J 期完工，滚动开工
- φ_j : 分期付款

$S_{j,t+1} = S_{j+1,t}$ （t期余j+1期完工的项目）

t期所有在建项目付款: $\sum_{j}^{J} \varphi_j \cdot S_{jt}$

$$\Rightarrow i_t = \sum_{j}^{J} \varphi_j \cdot S_{jt} + z_{t+1} - z_t$$

$$k_{t+1} = (1-\delta) k_t + S_{1t}$$

引入劳动衰减

$$U(C_t, h_t) = \frac{\left(C_t^{\frac{1}{3}} h_t^{\frac{2}{3}}\right)^{\gamma}}{\gamma} \qquad \gamma<1,\ \gamma\neq 0$$

$$h_t = \sum_{i=0}^{\infty} \alpha_i \cdot l_{t-i}$$ （过去的闲暇会影响当期效用，加权）

$$\begin{cases} \sum_i \alpha_i = 1 \\ \alpha_i = (1-\delta)^{i-1}\alpha_1 \\ \alpha_1 = \delta(1-\alpha_0) \end{cases}$$ （权重衰减）

$$\Rightarrow h_t = \sum_i \alpha_i (1-n_{t-i})$$

$$= \sum_i \alpha_i - \sum_i \alpha_i n_{t-i}$$

$$= 1 - \sum_i \alpha_i n_{t-i}$$

$$= 1-\alpha_0 n_t - \sum_{i=1}^{\infty} \alpha_i n_{t-i}$$

$$= 1-\alpha_0 n_t - \sum_{i=1} \alpha_1 (1-\delta)^{i-1} \cdot n_{t-i}$$

$$= 1-\alpha_0 n_t - (1-\alpha_0)\delta \boxed{\sum_{i=1} (1-\delta)^{i-1} \cdot n_{t-i}}$$

令 $\equiv a_t$

$$\Rightarrow a_{t+1} = (1-\delta)a_t + n_t$$

$$= 1-\alpha_0 n_t - (1-\alpha_0)\delta a_t$$

引入冲击.

真实冲击 $\lambda_t = \bar{\lambda} + \underbrace{u_t}_{\text{长期冲击}} + \underbrace{e_t}_{\text{短期冲击}}$

$$\begin{cases} u_t = \rho u_{t-1} + \underbrace{\varepsilon^1_{t-1}} \\ e_t = \underbrace{\varepsilon^2_{t-1}} \end{cases}$$

白噪

观测冲击: $\pi_t = \lambda_t + \varepsilon^3_t$

$$= \bar{\lambda} + u_t + e_t + \varepsilon^3_t$$

决策规则:

期初: 由 π_t 决定 n_t, S_{Jt}

$\pi_t \to \lambda_t$ 修正

期末: 由 λ_t 决策 C_t, Z_{t+1}

转为状态空间

$$\gamma_t = \begin{bmatrix} u_t \\ e_t \end{bmatrix}$$

$$\Rightarrow \gamma_{t+1} = \begin{bmatrix} \rho & 0 \\ 0 & 0 \end{bmatrix} \cdot \gamma_t + \begin{bmatrix} \varepsilon_t^1 \\ \varepsilon_t^2 \end{bmatrix}$$

$$\Rightarrow 1^{st}\text{观测} \quad P_t^1 = \pi_t - \bar{\pi} = [1, 1] \cdot \gamma_t + \underline{\varepsilon_t^3}$$

修正

$$\Rightarrow 2^{nd}\text{观测} \quad P_t^2 = \pi_t - \bar{\pi} = [1, 1] \cdot \gamma_t + \underline{0}$$

汇总：

$Var(\varepsilon_1)$

$$\begin{cases} \gamma_{t+1} = A\gamma_t + \epsilon_t^0, & A = \begin{bmatrix} \rho & 0 \\ 0 & 0 \end{bmatrix}, \quad \epsilon_t^0 = \begin{bmatrix} \varepsilon_t^1 \\ \varepsilon_t^2 \end{bmatrix} \sim N(0, V_0), \quad V_0 = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \\ P_t^1 = B\gamma_t + \epsilon_t^1, & B = [1, 1], \quad \epsilon_t^1 = \varepsilon_t^3, \quad \epsilon_t^1 \sim N(0, V_1), \quad V_1 = Var(\varepsilon^3) \\ P_t^2 = B\gamma_t + \epsilon_t^2, & B = [1, 1], \quad \epsilon_t^2 = 0 \end{cases}$$

	均值	协方差
基于上期	m^0	Σ^0
期初（1^{st}观测后）	m^1	Σ^1
期末（2^{nd}观测后）	m^2	Σ^2

使用 Kalman filter 更新.

stp 0：基于 m_{t-1}^2，Σ_{t-1}^2 预测

$$\begin{cases} m_t^0 = A \cdot m_{t-1}^2 \\ \Sigma_t^0 = A \cdot \Sigma_{t-1}^2 \cdot A + V_0 \end{cases}$$

stp 1：

$$\begin{cases} \hat{p}_t^1 = B \cdot m_t^0 & \text{（理论值）} \\ p_t^1 = B \cdot \gamma_t + \epsilon_t^1 & \text{（实际观测值）} \end{cases}$$

$$\Rightarrow V_t^1 = p_t^1 - \hat{p}_t^1$$

$$= B\gamma_t + \epsilon_t^1 - B m_t^0$$

$$= B \cdot \underbrace{(\gamma_t - m_t^0)}_{\text{系统运动偏差}} + \underbrace{\epsilon_t^1}_{\text{观测设备}}$$

$$\Rightarrow V_t^1 \sim N(0, S_1)$$

$$S_1 = B \Sigma_t^0 B + V_1$$

根据观测修正

$$K_1 = \Sigma_t^0 B S_1^{-1}$$ (1st的 Kalman Gains)

$$\Rightarrow \begin{cases} m_t^1 = m_t^0 + K_1 \cdot V_t^1 \\ \Sigma_t^1 = \Sigma_t^0 - K_1 B \Sigma_t^0 \end{cases}$$ 指导决策 η_t, S_{Jt}

step 2:

$$\begin{cases} \hat{P}_t^2 = B m_t^1 \\ P_t^2 = B \gamma_t + \epsilon_t^2 \end{cases}$$

$$\Rightarrow V_t^2 = B(\gamma_t - m_t^1)$$

$$\Rightarrow V_t^2 \sim N(0, S_2)$$

$$S_2 = B \cdot \Sigma_t^1 B$$

$$\Rightarrow K_2 = \Sigma_t^1 B S_2^{-1}$$

$$\Rightarrow \begin{cases} m_t^2 = m_t^1 + K_2 \cdot V_t^2 \\ \Sigma_t^2 = \Sigma_t^1 - K_2 B \cdot \Sigma_t^1 \end{cases}$$

求解均衡

Social Planner

$$\text{maximize: } E\cdot\sum_{t=0}^{\infty}\beta^t\cdot u\left[C_t,\ 1-\alpha_0 n_t-\mathcal{J}(1-\alpha_0)a_t\right]$$

$$\text{s.t.}\quad \begin{cases} k_{t+1}=(1-\delta)k_t+S_{1t} \\ S_{j\,t+1}=S_{j+1,t} \\ i_t=\sum_{j}^{J}\varphi_j\cdot S_{jt}+Z_{t+1}-Z_t \\ C_t+i_t=y_t \\ a_{t+1}=(1-\mathcal{J})a_t+n_t \\ \gamma_{t+1}=A\gamma_t+\epsilon_t^0 \\ P_t^1=B\gamma_t+\epsilon_t^1 \\ P_t^2=B\gamma_t+\epsilon_t^2 \end{cases}$$

$$\Rightarrow \begin{cases} n_t=n\left(k_t,\ S_{1t},\cdots S_{J-1,t},\ Z_t,\ a_t,\ m_t^1\right) \\ S_{Jt}=S\left(k_t,\ S_{1t},\cdots S_{J-1,t},\ Z_t,\ a_t,\ m_t^1\right) \\ C_t=C\left(k_t,\ S_{1t}\cdots S_{J-1,t},\ S_{Jt},\ Z_t,\ a_t,\ n_t,\ m_t^2\right) \\ Z_{t+1}=Z\left(k_t,\ S_{1t}\cdots S_{J-1,t},\ S_{Jt},\ Z_t,\ a_t,\ n_t,\ m_t^2\right) \end{cases}$$

Steady State.

HH:

$$u'(C_t) = \beta(1+r_t)\cdot u'(C_{t+1})$$

$$\Rightarrow u'(\bar{C}) = \beta\cdot(1+\bar{r})\cdot u'(\bar{C})$$

$$\Rightarrow \bar{r} = \frac{1-\beta}{\beta}$$

资本建造.

$$q = \varphi_J\cdot(1+r)^{J-1} + \varphi_{J-1}\cdot(1+r)^{J-2} + \cdots + \varphi_1\cdot(1+r)^1 = \sum_{j=1}^{J}\varphi_j\cdot(1+r)^{j-1}$$

(t期一份资本的建造成本)

$$q(1+r) = f_k + (1-\delta)\cdot q$$

$$\Rightarrow f_k = (r+\delta)\cdot q$$

存货

$$1\cdot(1+r) = f_z + 1$$

$$\Rightarrow f_z = r$$

$$\Rightarrow \frac{f_k}{f_z} = \frac{q(r+\delta)}{r}$$

由于 $f = A\cdot n^{\theta}\cdot S^{1-\theta}$

$$S = \left[(1-\sigma)k^{-v} + \sigma\cdot z^{-v}\right]^{\frac{-1}{v}}$$

$$\Rightarrow \begin{cases} f_k = A\cdot n^{\theta}\cdot(1-\theta)\cdot S^{-\theta}\cdot\frac{-1}{v}\cdot\left[\quad\right]^{\frac{-1}{v}-1}(1-\sigma)\cdot(-v)\cdot k^{-v-1} \\ f_z = A\cdot n^{\theta}\cdot(1-\theta)\cdot S^{-\theta}\cdot\frac{-1}{v}\cdot\left[\quad\right]^{\frac{-1}{v}-1}\cdot\sigma\cdot(-v)\cdot z^{-v-1} \end{cases}$$

$$\Rightarrow \frac{f_k}{f_z} = \frac{1-\sigma}{\sigma}\cdot\left(\frac{k}{z}\right)^{-v-1} = \frac{q(r+\delta)}{r}$$

$$\Rightarrow z = \overbrace{\boxed{\left[\frac{q(r+\delta)}{r}\cdot\frac{\sigma}{1-\sigma}\right]^{\frac{1}{v+1}}}}^{\text{令 } b_1}\cdot k$$

$$\Rightarrow Z = b_1 \cdot k$$

$$\Rightarrow S = \left[(1-\sigma)k^{-v} + \sigma \cdot b_1^{-v} \cdot k^{-v}\right]^{\frac{-1}{v}}$$

$$= \left[\overbrace{\boxed{\underline{(1-\sigma+\sigma \cdot b_1^{-v})}}}^{\triangleq\ b_2} \cdot k^{-v}\right]^{\frac{-1}{v}}$$

$$= b_2^{\frac{-1}{v}} \cdot k$$

$$\Rightarrow f_k = \lambda \cdot n^{\theta} (1-\theta) \cdot S^{-\theta} \cdot \frac{-1}{v}\left(S^{-v}\right)^{\frac{-1-v}{v}} \cdot (1-\sigma) \cdot (-v) \cdot k^{-v-1} = q \cdot (r+\delta)$$

$$\Leftrightarrow \lambda \cdot n^{\theta} \cdot (1-\theta) \cdot S^{v+1-\theta} \cdot (1-\sigma) \cdot k^{-v-1} = q \cdot (r+\delta)$$

$$\Leftrightarrow \lambda \cdot n^{\theta} \cdot (1-\theta)\left(b_2^{\frac{-1}{v}} \cdot k\right)^{v+1-\theta} \cdot k^{-v-1} = q \cdot (r+\delta)$$

$$\Leftrightarrow \lambda \cdot n^{\theta} (1-\theta) \cdot (1-\delta) \cdot b_2^{\frac{\theta-1-v}{v}} \cdot k^{-\theta} = q\,(r+\delta)$$

$$\Rightarrow k = \left[\frac{\lambda(1-\theta)(1-\sigma) \cdot b_2^{\frac{\theta-1-v}{v}}}{q(r+\delta)}\right]^{\frac{1}{\theta}} \cdot n$$

令 $b_3 = \left[\dfrac{(1-\theta)(1-\sigma)\cdot b_2^{\frac{\theta-1-v}{v}}}{q(r+\delta)} \right]^{\frac{1}{\theta}}$

$$\Rightarrow k = \lambda^{\frac{1}{\theta}} \cdot b_3 \cdot n$$

$$\Rightarrow z = b_1 b_3 \lambda^{\frac{1}{\theta}} \cdot n$$

故产出

$$f = \lambda n^{\theta} \cdot s^{1-\theta}$$

$$= \lambda \cdot n^{\theta} \cdot \left(b_2^{\frac{-1}{v}} \cdot k\right)^{1-\theta}$$

$$= \lambda \cdot n^{\theta} \cdot \left(b_2^{\frac{-1}{v}} \cdot b_3 n \lambda^{\frac{1}{\theta}}\right)^{1-\theta}$$

$$= \lambda^{\frac{1}{\theta}} \cdot \overbrace{\boxed{b_2^{\frac{\theta-1}{v}} \cdot b_3^{1-\theta}}}^{\text{令 } b_4} \cdot n$$

$$= b_4 \cdot \lambda^{\frac{1}{\theta}} \cdot n$$

市场出清:

$$f = C + i = C + \delta \cdot k$$

$$\Rightarrow C = b_4 \cdot \lambda^{\frac{1}{\alpha}} \cdot n - \delta b_3 n \lambda^{\frac{1}{\alpha}}$$

$$= (b_4 - b_3 \cdot \delta) \cdot \lambda^{\frac{1}{\alpha}} \cdot n$$

Summary
$$\begin{cases} k = b_3 \cdot \lambda^{\frac{1}{\alpha}} \cdot n \\ z = b_1 b_3 \lambda^{\frac{1}{\alpha}} n \\ f = b_4 \lambda^{\frac{1}{\alpha}} \cdot n \\ C = (b_4 - \delta b_3) \lambda^{\frac{1}{\alpha}} \cdot n \end{cases}$$

对HH

$$U = \frac{\left[C_t^{\frac{1}{3}} \cdot h_t^{\frac{2}{3}}\right]^{\gamma}}{\gamma}$$

令 μ: 影子价格, W: 工资

对 C_t:

$$\left(C_t^{\frac{1}{3}} \cdot h_t^{\frac{2}{3}}\right)^{\gamma-1} \cdot h_t^{\frac{2}{3}} \cdot C_t^{\frac{2}{3}} \cdot \frac{1}{3} = \mu$$

$$\Rightarrow \frac{1}{3} \cdot C^{\frac{\gamma}{3}-1} \cdot h^{\frac{2\gamma}{3}} = \mu$$

对 l_t:

工作带来的效用: $\beta^t \cdot \mu_t \cdot W_t$

闲暇的效用:

$$t: \frac{\partial u_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial l_t} \cdot \beta^t$$

$$t+1: \frac{\partial u_{t+1}}{\partial h_{t+1}} \cdot \frac{\partial h_{t+1}}{\partial l_t} \cdot \beta^{t+1}$$

$$\vdots$$

$$= \sum_{i=0}^{\infty} \frac{\partial u_{t+i}}{\partial h_{t+i}} \cdot \frac{\partial h_{t+i}}{\partial l_t} \cdot \beta^{t+i}$$

$$= \beta^t \cdot \sum_i \beta^i \cdot C_{t+i}^{\frac{\gamma}{3}} \cdot h_{t+i}^{\frac{2\gamma}{3}-1} \cdot \frac{2}{3} \cdot \alpha_i$$

$\Rightarrow$ 均衡：$\beta^t \cdot \mu_t \cdot w_t = \beta^t \cdot \sum_i \beta^i \cdot C_{t+i}^{\frac{\gamma}{3}} \cdot h_{t+i}^{\frac{2\gamma}{3}-1} \cdot \frac{2}{3} \cdot \alpha_i$

$\Rightarrow$ 稳态：$\mu \cdot w = \frac{2}{3} \cdot \sum_i^{\infty} \beta^i \cdot C^{\frac{\gamma}{3}} \cdot h^{\frac{2\gamma}{3}-1} \cdot \alpha_i$

由于 $h_t = \sum_i \alpha_i \cdot l_{t-i} \Rightarrow h = \sum_i \alpha_i \cdot l = l$

$$\Rightarrow \begin{cases} \mu = \frac{1}{3} \cdot C^{\frac{\gamma}{3}-1} \cdot l^{\frac{2\gamma}{3}} \\ \mu \cdot w = \frac{2}{3} C^{\frac{\gamma}{3}} \cdot l^{\frac{2\gamma}{3}-1} \cdot \sum_i \beta^i \cdot \alpha_i \end{cases}$$

$\vdots$

$$w \cdot (1-n) = 2C \cdot \left[\alpha_0 + \frac{(1-\alpha_0) \cdot \eta}{r+\eta} \right]$$